# The Causal Effect of Old Age Security on Labor Force Exit

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## Introduction

The transition from the labour force into retirement represents one of the most significant shifts in an individual's economic lifecycle. From the perspective of microeconomic theory, this transition is often viewed through the lens of the trade off between labour and leisure. Standard models predict that a sudden, exogenous increase in non-labour income like government pension should increase the demand for leisure, thereby leading to a sharp decline in the Labour Force Participation Rate (LFPR). In Canada, this theoretical "income effect" is at the age of 65.

The incentives driving retirement in Canada can be viewed through the 3 pillars of retirement income as described by Ambrozas et al. (2020). Pillar 1, the Old Age Security (OAS) is considered fundamental because it is available to all eligible individuals regardless of their prior work history or individual contribution to public systems, Pillar 2 represents the Canada or Quebec pension plan, a work related program intended to replace approximately 33% of a worker's income. The last pillar, pillar 3, includes private pensions and personal savings, which can vary significantly depending on the individual's workplace benefits. While all three pillars contribute to the decision to retire, this paper focuses on specifically the first pillar, OAS. The OAS is intended for medium to low income individuals aged 65 and up that is a Canadian citizen or a legal resident who has lived in Canada for at least 10 years after the age of 18.

The motivation for this study comes from the concerns about the fiscal sustainability of Canada's aging population. According to the 18th Actuarial Report on the Old Age Security Program (Office of the Chief Actuary, 2023), the number of OAS beneficiaries is projected to grow from 7.2 million in 2023 to 9.8 million by 2035. Simultaneously the ratio of the number of individuals aged 20-65 for every senior aged 65+ is also expected to fall from 43.1 in 2023 to 2.1 by 2060. This demographic shift creates a challenge of

increasing government expenditure while shrinking the government revenue from taxes. If OAS eligibility functions as a causal driver for retirement, the program may be inadvertently distorting labor supply decisions, leading to an earlier exit from the workforce than would occur in the absence of benefits otherwise.

Therefore, this paper aims to address this issue by determining: W*hat is the causal effect of OAS eligibility at age 65 on the probability of exiting the labour force for older adults in Canada?* To isolate this effect, I use a Difference-in-Differences (DiD) method, with the 10 year residency rule to compare eligible residents against ineligible recent immigrants.

## Data

The analysis uses data from the 2021 Canadian Census PUMF. To focus strictly on the transition around the age 65 threshold, the sample is restricted to individuals between the ages of 60-65. Non permanent residents are excluded from the sample as they are subject to distinct labor and immigration regulations. In addition, the sample also excludes individuals who reported having "never worked" in their lifetime. The restriction ensures that the analysis measures genuine labor force exit among individuals with a history of involvement in the labor force.

**Outcome:**

The primary outcome variable is the *NILF*, which is encoded as a binary dummy variable that equals 1 if the individual is neither employed or actively seeking work, and 0 if they are currently employed or officially unemployed but actively seeking work.

**Treatment:**

The analysis relies on determining an individual's legal eligibility for Old Age Security (OAS). Because the Census does not have a direct "OAS Eligible" flag, this was constructed using two raw Census variables:

- Immigration Status (*immstat*): Used to separate domestic-born Canadian citizens from immigrants.
- Year of Immigration (*yrim*): Used to determine how long an immigrant has resided in Canada.

These variables were combined to create a binary Eligible dummy variable. It equals 1 for individuals legally permitted to receive OAS (domestic-born Canadians, or immigrants who arrived in 2011 or earlier, satisfying the strict 10-year residency rule). It equals 0 for recent immigrants who arrived after 2011 and are not allowed to receive the pension.

**Covariates:**

- Education (*educ_level*): A four-level categorical variable (Less than High School, High School, College/Trades, University) that is representative of an individual's lifetime earning trajectory.
- Marital Status (*marital_group*): A categorical variable (Partnered, Previously Married, Single) that captures joint household retirement decisions, economies of scale in living expenses, and dual income wealth pooling.
- Gender (*female*): A binary indicator controlling for historical, structural differences in lifetime labor force attachment and primary household earner dynamics for this specific generational cohort.
- Visible Minority Status (*vismin_dummy*): A binary indicator used to control for systemic wage disparities in the Canadian labor market. Because lower lifetime earnings mathematically result in lower CPP payouts and diminished private

savings, this variable controls for the structural necessity to remain in the labor market past age 65 independently of OAS eligibility.

Table 1—Summary Statistics of Key Variables

| Variable | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|
| Not in Labour Force (NILF) | 0.540 | 0.498 | 0 | 1 |
| Eligible for OAS | 0.987 | 0.114 | 0 | 1 |
| Age 65-69 | 0.454 | 0.498 | 0 | 1 |
| Female | 0.509 | 0.500 | 0 | 1 |
| Visible Minority | 0.145 | 0.352 | 0 | 1 |
| Education: Less Than High School | 0.149 | 0.356 | 0 | 1 |
| Education: High School | 0.298 | 0.457 | 0 | 1 |
| Education: College or Trades | 0.340 | 0.474 | 0 | 1 |
| Education: University | 0.213 | 0.409 | 0 | 1 |
| Marital: Partnered | 0.714 | 0.452 | 0 | 1 |
| Marital: Previously Married | 0.195 | 0.396 | 0 | 1 |
| Marital: Never Married | 0.091 | 0.287 | 0 | 1 |

## Methodology

To start, a simple OLS regression is used comparing those eligible for OAS to those who are not.

(1) $$NILF_i = \alpha + \beta_1 eligible_i + \epsilon_i$$

To interpret $\beta_1$ as a causal effect independent of bias where $\beta_1 = ATT + Bias$, the independence assumption must be satisfied. This means eligibility is random and *E[Yi(0)|Di=1]=E[Yi(0)|Di=0]*. In this context, this implies that in the absence of treatment, both groups would have the same probability of retiring. However, this baseline assumption is too strong and likely violated in observational data due to selection bias.

To test this, a two sample t test on the covariate *vismin_dummy* (visible minority status) will be used. Visible minority status serves as a critical proxy for systemic wage differentials and lifetime wealth accumulation. If the null hypothesis of equality is rejected, it proves that the treatment and control groups are structurally different, rendering the naive $\beta 1$ biased. The exact null and alternative hypothesis are as follows:

*H0*: There is no statistically significant difference in the proportion of visible minorities between *eligible* and *ineligible* groups.

*H1*: There exists a statistically significant difference in the proportion of visible minorities between *eligible* and *ineligible* groups.

TABLE 2—RESULTS OF BALANCE TEST FOR VISIBLE MINORITY STATUS

| Balance test: Visible minority by eligibility | | | | | | |
|---|---|---|---|---|---|---|
| Mean (Eligible=0) | Mean (Eligible=1) | Diff (1-0) | t-stat | p-value | CI Low | CI High |
| 0.818 | 0.136 | 0.682 | 68.464 | 0 | 0.663 | 0.702 |

With a p-value of < 0.01, we reject the null hypothesis in favor of the alternative hypothesis. This confirms the presence of selection on observables (*B2 bias*), necessitating the addition of the full vector of control covariates: *education*, *marital status*, *gender*, and *visible minority status*.

(2) $$NILF_i = \alpha + \beta_1 eligible_i + \tau X_i + \epsilon_i$$

This follows the conditional Independence assumption, $E[Y_i(0)|D_i=1, X_i=x]=E[Y_i(0)|D_i=0, X_i=x]$. This assumes that once we control for the observable traits $X_i$, eligibility is as good as random. However, if there are unobservable differences, such as undocumented private savings, this model would still suffer from *B3 bias*, also known as selection on unobservables.

To control for these time invariant unobservables we use a difference in differences framework. By using *agegrp* (60-64, 65-69), we can difference out the baseline traits of domestic Canadians and recent immigrants.

(3) $$NILF_i = \alpha + \beta_1 eligible_i + \beta_2 agegrp + \beta_3 (eligible \times agegrp) + \epsilon_i$$

The validity of the DiD depends on the parallel trends assumption, which points that both groups would follow the same retirement trajectory in the absence of the OAS benefit. Based on the compositional differences established by the t-test, this assumption could be threatened. In this case, I estimate a Full DiD Model that includes the interaction of covariates to allow for heterogeneous trends across demographic groups:

(4) $NILF_i = \alpha + \beta_1 eligible_i + \beta_2 agegrp + \delta(eligible \times agegrp) + \tau Xi + \epsilon_i$

The coefficient  is the parameter of interest, representing the causal impact of OAS eligibility on the probability of labour force exit.

## Results

TABLE 3—REGRESSION RESULTS OF OAS ELIGIBILITY ON LABOUR FORCE EXIT

Cross-sectional DD-style models

| | Dependent variable: | | | |
|---|---|---|---|---|
| | Not in labour force (NILF) | | | |
| | Baseline (1) | Controls (2) | DiD (3) | DiD + controls (4) |
| Eligible (OAS) | 0.015 | -0.038*** | -0.005 | -0.062*** |
| | (0.013) | (0.013) | (0.017) | (0.017) |
| Age 65-69 | | 0.288*** | 0.238*** | 0.229*** |
| | | (0.003) | (0.025) | (0.024) |
| Female | | 0.115*** | | 0.115*** |
| | | (0.003) | | (0.003) |
| High school (ref: < HS) | | -0.050*** | | -0.050*** |
| | | (0.004) | | (0.004) |
| College/Trades (ref: < HS) | | -0.068*** | | -0.068*** |
| | | (0.004) | | (0.004) |
| University (ref: < HS) | | -0.106*** | | -0.106*** |
| | | (0.005) | | (0.005) |
| Previously married (ref: partnered) | | -0.010*** | | -0.010*** |
| | | (0.004) | | (0.004) |
| Never Married (ref: partnered) | | 0.055*** | | 0.055*** |
| | | (0.005) | | (0.005) |
| Visible minority | | -0.088*** | | -0.088*** |
| | | (0.004) | | (0.004) |
| Eligible x Age 65-69 | | | 0.050** | 0.052** |
| | | | (0.025) | (0.024) |
| Constant | 0.526*** | 0.461*** | 0.419*** | 0.485*** |
| | (0.013) | (0.013) | (0.017) | (0.017) |
| Observations | 115,096 | 115,096 | 115,096 | 115,096 |
| R2 | 0.00001 | 0.099 | 0.078 | 0.099 |
| Adjusted R2 | 0.00000 | 0.099 | 0.078 | 0.099 |
| Residual Std. Error | 0.498 (df = 115094) | 0.473 (df = 115086) | 0.479 (df = 115092) | 0.473 (df = 115085) |
| F Statistic | 1.291 (df = 1; 115094) | 1,411.634*** (df = 9; 115086) | 3,247.100*** (df = 3; 115092) | 1,270.958*** (df = 10; 115085) |
| Note: | | | | *p<0.1; **p<0.05; ***p<0.01 |

In *column 1*, a simple difference in means shows a statistically insignificant 1.5 percentage point difference in retirement rates between the *eligible* and *ineligible* groups.

However, this estimate is heavily biased, as established by the hypothesis test, as it fails to account for any confounding factors.

In *column 2*, this model adds demographic controls and the age dummy. With this, we can be 99% confident that aging into the 65–69 bracket is associated with a 28.0 percentage point increase in the probability of being out of the labor force, holding all else constant. However, the -3.8 percentage points for *Eligible* (OAS) demonstrates that a pooled model is insufficient as it incorrectly assumes the effect of being in the *Eligible* group is constant across all ages, failing to isolate the specific jump at age 65.

In *column 3*, we use the DiD model. The model isolates the pure effect of aging from OAS. We are 99% confident that going from the 60–64 to 65–69 causes a 23.0 percentage point drop in labour force participation, *ceteris paribus* and completely independent of OAS. While all individuals show a higher probability of retirement after age 65, the availability of the OAS pension triggers an additional 5.0 percentage point increase in labour force exit specifically for the *eligible* group.

*Column 4* represents the preferred specification because it accounts for the demographic imbalances identified in the initial selection bias tests. Given the compositional differences between the two groups, it is unlikely their labor market trajectories would be naturally parallel. By incorporating covariates, the model shifts to a conditional parallel trends assumption, meaning we assume the groups would trend together once we hold covariates constant. Under these controls, we are 95% confident that OAS *eligibility* is associated with a 5.2 percentage point increase in the probability of exiting the labour force.

It is also worth looking at the covariates for the preferred specification. All of them are statistically significant at the 99% level. Holding all other factors constant, *women* in this

cohort are 11.5 percentage points more likely to be out of the labor force than men. This could be because women often had more interrupted career paths due to caregiving responsibilities. This results in lower accumulated Canada Pension Plan (CPP) benefits and smaller private pensions, which might ironically lead to earlier labor force exit if they rely on a partner's higher retirement income. Meanwhile, individuals with a *University degree* are 10.6 percentage points less likely to be retired than those without a high school diploma. This aligns with the theory that higher human capital leads to higher opportunity costs for leisure and typically involves less physically demanding labor, both of which delay retirement. Holding all else constant, *visible minorities* are 8.0 percentage points less likely to be retired than non-minorities. This coefficient likely captures the systemic earning disparities that result in lower Canada Pension Plan (CPP) accumulation and diminished private savings, structurally forcing these populations to remain in the labor force longer to maintain subsistence income. Individuals who were *Never Married* are 5.5 percentage points less likely to be retired than those who are partnered. This reflects the loss of "economies of scale" in household expenses and the lack of pooled retirement assets, increasing the individual financial pressure to continue working.

Ultimately, this model answers the causal question by proving that even when we compare individuals with the exact same *education*, *gender*, and *visible minority status*, becoming *eligible* for OAS is still associated with a 5.2 percentage point increase in the likelihood of leaving the workforce. This confirms that the pension is a primary driver of retirement behavior in Canada.

## Discussion and Extension

The final model relies on the conditional parallel trends assumption, where once we control for *education*, *gender*, and *visible minority status*, their underlying retirement

trends of the domestic and immigrant groups would have been parallel in the absence of the OAS policy.

We observe that domestic Canadians retire at a significantly higher rate after age 65. The DiD model uses the recent immigrant group as a counterfactual. This allows us to determine what would happen to domestic Canadians if OAS was taken away. Because this "jump" happens exactly at the legal eligibility threshold and only for the group that is legally allowed to receive the money, we can interpret this as a causal liquidity effect. The sudden arrival of non labor income (OAS) reduces the marginal utility of work, thus leading to an immediate shift into retirement.

A possible extension could be using fuzzy regression discontinuity using year of immigration as the running variable. While the current model relies on the conditional parallel trends assumption, a Fuzzy RD would only require that individuals arriving in Canada just before or just after the 10-year cutoff are locally identical in their unobservable characteristics. By using this cutoff as an Instrumental Variable for OAS, we could isolate the Local Average Treatment Effect (LATE). Crucially, to ensure the resulting estimate is a mathematically valid weighted average of these local effects, the model would require a saturated set of covariates and their interactions with the instrument.

By ensuring that all weights remain positive and interpretable, this fuzzy regression discontinuity model would help with concerns about selection bias between domestic Canadians and immigrants, as the comparison would be narrowed down to immigrants who are virtually identical except for one year of arrival time. If this Fuzzy RD yielded a similar 5.2 percentage point effect, it would provide further evidence that OAS is the driving force for the labour force exit.

## Conclusion

This paper demonstrates that Old Age Security (OAS) eligibility at age 65 has a statistically significant causal impact on retirement behavior. While the natural aging process accounts for a 23.0 percentage point increase in labour force exit, becoming legally eligible for the pension is associated with an additional 5.2 percentage point increase in the probability of being out of the labour force. Ultimately, these results carry significant implications for the fiscal sustainability concerns raised in the introduction. As the dependency ratio of workers to seniors is projected to fall to 2.1 by 2060, the identified 5.2 percentage point effect proves that OAS eligibility inadvertently distorts labor supply decisions (Office of the Chief Actuary, 2023). By providing a financial incentive for individuals to exit the workforce earlier than they might in a benefit free counterfactual, the program may be exacerbating the challenge of shrinking tax revenues and increasing government expenditures.

Outside of using fuzzy regression discontinuity, I would consider adding more covariates to control for household wealth and regional policy variations. One such covariate could be province, as there could be differences between CPP and QPP as well as regional cost of living disparities leading to different impacts depending on the province. Additionally, Housing Tenure (renter vs homeowner) could be a crucial indicator of household wealth and financial stability. By refining the model with these additional controls, the analysis could more precisely distinguish the pure policy effect of OAS from the broader financial and familial pressures that define the Canadian retirement landscape.

## References

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